

Characteristic-Space Metrics in Factor Models: Identification and Inference

Mingyang Liu*

September 2026

Abstract

Characteristic-based factor models can fit returns identically while attributing them differently to latent, observed-factor, and intercept components. We show how an inner product on characteristic space selects this decomposition and formulate its loading class modulo latent rotations. For an exactly profiled estimator with deterministic date weights and an estimated Gram metric, we derive a linear representation retaining both return-score and metric uncertainty. Smooth reports admit chart-invariant inference and feasible long-run covariance estimation. Covariance alignment provides sufficient conditions for structural interpretation with a fixed cross section. Near a smooth exposure boundary, metric estimation also moves the feasible set. Joint Gaussian projection yields drift-robust scalar confidence intervals. For one constraint, the least-favorable tail envelopes reduce to the interior and binding endpoints. Numerical experiments distinguish omitted metric uncertainty from omitted boundary movement.

Keywords: characteristic-based factor models; metric-indexed identification; estimated Gram metrics; rotation-invariant inference; inference under inequality constraints.

JEL Classification: C13, C38, C58, G12.

*Imperial Business School, Imperial College London.

1 Introduction

A factor model can reproduce returns without uniquely allocating them to its components. When latent factors coexist with observed factors and a characteristic-dependent intercept, a change in the latent realizations can offset a reallocation of their loadings. The complete fitted index is unchanged, but its decomposition is not. This distinction matters whenever researchers compare benchmark exposures, measure a residual intercept, or ask which characteristic directions the latent block spans. Residual fit evaluates the complete index; it cannot resolve an ambiguity that leaves that index unchanged.

The allocation uses an inner product on characteristic space. Firm characteristics differ in scale and overlap, even after marginal standardization. Euclidean orthogonality compares their recorded coefficients. A characteristic Gram metric instead compares the cross-sectional exposure patterns those coefficients generate. We make this choice explicit and study its consequences for identification and inference. The metric specifies a convention for separating loading blocks; a population criterion and positive local curvature identify the target within that convention.

The paper has three results. First, we construct an exact matched transformation for any positive-definite metric. It normalizes the latent loadings and makes the retained blocks orthogonal to the latent span, while an offsetting factor transformation preserves the fitted index at every date. Within this transformation orbit, only orthogonal rotations of the normalized latent block remain. A finite bound on intercept exposure has a different role: it restricts the feasible set and can change the fitted index itself.

Second, we derive inference for a selected local minimizer of an exactly profiled, fixed-weight criterion when the Gram metric is estimated from the same panel. Local quotient coordinates remove latent rotations. The resulting root- T linear representation combines the profiled return score with metric-estimation variation. A smooth report of the loading class inherits both channels, including their cross-covariance. Its influence function and asymptotic covariance are unchanged by an admissible change of chart. This permits inference for latent-span alignment, benchmark-span distance, and retained intercept exposure without interpreting arbitrary latent columns.

The time dimension supplies the asymptotic index; the cross section need not diverge. Consequently, the population target of the profiled criterion need not equal the structural loading class in the return equation. Profiling uses the same disturbance to estimate each date's factors and form its loading score. We isolate the resulting covariance term and give a

sufficient covariance-alignment condition for structural and criterion targets to coincide. This condition supplies a structural interpretation of the inference without requiring date-specific factor errors to vanish.

Third, we study one smooth exposure inequality near binding. The learned metric then perturbs the score, shifts the local feasible boundary, and may enter the report directly. A joint Gaussian limit followed by a curvature-weighted projection captures these effects. Least-favorable quantiles yield potentially conservative intervals uniformly over feasible local drifts from strict slackness to binding. For the scalar, single-half-space problem studied here, a common-draw monotonicity argument reduces the quantile envelope to its two endpoints. Uniform inference therefore does not require estimating local slack or searching a growing drift grid.

Our analysis complements the factor-model literature. Bai (2003); Bai and Ng (2013) study identification and inference in approximate factor models; Kelly et al. (2019) use characteristics to instrument time-varying loadings. The joint large- N , large- T theory of Kelly et al. (2026) concerns recovery of structural loadings and factors under normalization. Here the object is a metric-indexed loading class with retained labels, and the fixed-dimensional root- T theory allows a fixed cross section. The distinction is consequential: resolving rotations does not by itself remove the effect of factor profiling on the population target.

The methods build on matrix-manifold geometry (Edelman et al., 1998; Absil et al., 2008), two-step extremum estimation (Newey, 1994; Newey and McFadden, 1994), and constrained asymptotics (Geyer, 1994; Andrews, 1999). Least-favorable inference is developed in Andrews and Guggenberger (2009); Andrews and Soares (2010). Li (2025) develops a proximal bootstrap for constrained estimators, including settings with estimated constraints. Our focus is the coupled score–boundary–report law when the same estimated metric selects a decomposition and its exposure constraint. The framework makes these dependencies explicit and carries them through feasible inference.

The numerical evidence follows the same sequence. An observed panel of 153 firm characteristics shows how many named signals can produce overlapping stock exposures. A known-truth factor-panel experiment then holds the point estimate fixed and isolates the effect of metric uncertainty on standard errors and confidence intervals. A separate local quadratic experiment measures the coverage and interval-length consequences of learning an exposure boundary. The accompanying figures display the geometry, the standardized estimation errors, and the transition to binding; the tables report the corresponding numerical comparisons. Together they show when a researcher would draw different conclusions

from the same estimated decomposition after accounting for how its measurement rule was learned.

Section 2 defines the model and metric. Section 3 constructs the quotient estimand. Sections 4–6 establish estimation and inference. Section 7 gives the learned-boundary theory, and Section 8 presents numerical evidence. Essential conditions and proof arguments are included in the main text; the Online Supplement provides derivative calculations, implementation details, and additional simulation results.

2 Model, Metric, and Quotient Estimand

At dates $t = 1, \dots, T$, observe N_t units with predetermined characteristics $z_{i,t} \in \mathbb{R}^L$ and subsequent outcomes $r_{i,t+1}$. In the asset-pricing application, units are firms and outcomes are excess returns. The theory takes $T \rightarrow \infty$, holding the numbers of characteristics and factors fixed; N_t may vary and need not diverge. The subscript $t + 1$ identifies the return period, not a different cross section. Stack $r_{t+1} \in \mathbb{R}^{N_t}$ and $Z_t = (z_{1,t}, \dots, z_{N_t,t})' \in \mathbb{R}^{N_t \times L}$. The model separates three sources of fitted variation. Its metric specifies how their loading directions are compared; its exposure budget limits the part retained outside the factor blocks.

The characteristic-based factor model is

$$r_{i,t+1} = z'_{i,t} \Gamma_\beta f_{t+1} + z'_{i,t} \Gamma_\delta g_{t+1} + z'_{i,t} \Gamma_\alpha + \varepsilon_{i,t+1}. \quad (2.1)$$

Here $\varepsilon_{i,t+1}$ is the unit-level disturbance, $f_{t+1} \in \mathbb{R}^K$ contains latent realizations, $g_{t+1} \in \mathbb{R}^{K_o}$ contains observed common components, $\Gamma_\beta \in \mathbb{R}^{L \times K}$ and $\Gamma_\delta \in \mathbb{R}^{L \times K_o}$ are loading maps, and $\Gamma_\alpha \in \mathbb{R}^L$ is the intercept loading. We assume $1 \leq K < L$; $K_o = 0$ omits the observed block. A superscript $\circ$ will distinguish structural objects in this equation from criterion targets.

Define the complete characteristic index and managed moments by

$$h_{t+1} = \Gamma_\beta f_{t+1} + \Gamma_\delta g_{t+1} + \Gamma_\alpha, \quad r_{t+1} = Z_t h_{t+1} + \varepsilon_{t+1}, \quad (2.2)$$

$$X_{t+1} = N_t^{-1} Z_t' r_{t+1}, \quad W_t = N_t^{-1} Z_t' Z_t. \quad (2.3)$$

Thus $X_{t+1} = W_t h_{t+1} + u_{t+1}$, where $u_{t+1} = N_t^{-1} Z_t' \varepsilon_{t+1}$ is the managed disturbance. The population Gram metric and its sample counterpart are

$$Q_Z = \mathbb{E} W_t \succ 0, \quad \widehat{Q}_Z = T^{-1} \sum_{t=1}^T W_t. \quad (2.4)$$

Individual W_t may be singular even when Q_Z is positive definite.

For loading directions $a, b \in \mathbb{R}^L$,

$$\langle a, b \rangle_{Q_Z} = a' Q_Z b = \mathbb{E} \left[N_t^{-1} (Z_t a)' (Z_t b) \right], \quad \|a\|_{Q_Z}^2 = a' Q_Z a. \quad (2.5)$$

The Gram metric measures overlap between the exposure patterns generated by characteristics, rather than between their coefficient vectors alone. It is an uncentered second moment: if $\bar{z}_t$ and $\Sigma_{z,t}^c$ are the within-date feature mean and centered covariance, then $W_t = \Sigma_{z,t}^c + \bar{z}_t \bar{z}_t'$. Size, value, momentum, profitability, and related transformations can therefore remain overlapping directions after individual standardization. Declaring the metric makes block-level comparisons reproducible and gives intercept exposure a measurable scale.

For $Q \succ 0$ and $\delta_\alpha \in [0, \infty]$, impose

$$\mathcal{M}(Q, \delta_\alpha) = \left\{ (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha) : \begin{array}{l} \Gamma_\beta' Q \Gamma_\beta = I_K, \quad \Gamma_\beta' Q [\Gamma_\delta, \Gamma_\alpha] = 0, \\ \Gamma_\alpha' Q \Gamma_\alpha \leq \delta_\alpha \end{array} \right\}. \quad (2.6)$$

The equalities normalize the latent block and separate it from retained labelled and intercept loadings. Section 3 shows that they select a representative through a matched transformation. The scalar δ_α is the declared intercept-exposure budget; $\delta_\alpha = \infty$ leaves that exposure unrestricted. Unlike the equalities, the inequality can exclude loading classes. At the population Gram metric,

$$\Gamma_\alpha' Q_Z \Gamma_\alpha = \mathbb{E} \left[N_t^{-1} \sum_{i=1}^{N_t} (z_{i,t}' \Gamma_\alpha)^2 \right]. \quad (2.7)$$

It bounds the mean-square characteristic-indexed intercept. In an asset-pricing application, this makes explicit how much fitted return exposure is left outside the latent and observed-factor blocks. The budget provides a controlled way to study how restricting the retained intercept changes the fitted decomposition. This connects the model comparison to conditional pricing questions arising from omitted factors and misspecified dynamics (Zaffaroni, 2025). Equation (2.7) anchors that comparison in the distribution of observed characteristics. A covariance-weighted pricing-error interpretation becomes available when a specified pricing model establishes the correspondence between this Gram-based exposure measure and pricing errors relative to its factor benchmark.

Orthogonal rotations remain observationally irrelevant. For $\mathcal{O}(K) = \{R \in \mathbb{R}^{K \times K} : R'R =$

$I_K\}$, replacing (Γ_β, f_{t+1}) by $(\Gamma_\beta R, R' f_{t+1})$ preserves their product. Write $\vartheta = (\Gamma_\beta, \Gamma_\delta, \Gamma_\alpha)$ and define

$$[\vartheta] = \{(\Gamma_\beta R, \Gamma_\delta, \Gamma_\alpha) : R \in \mathcal{O}(K)\} \in \mathcal{M}(\mathcal{Q}, \delta_\alpha) / \mathcal{O}(K). \quad (2.8)$$

The quotient is the collection of loading classes after removing latent rotations, not a numerical division. At a full-rank interior point its dimension is

$$d = (L - K)(K + K_o + 1). \quad (2.9)$$

A useful report of the latent span is the metric-whitened projector

$$\mathcal{P}_Q(\Gamma_\beta) = Q^{1/2} \Gamma_\beta (\Gamma_\beta' Q \Gamma_\beta)^{-1} \Gamma_\beta' Q^{1/2}, \quad (2.10)$$

where $Q^{1/2}$ is the symmetric positive-definite square root. This projector is invariant to the latent basis, but generally depends on Q . Inference below concerns such defined features of the loading class. The next section first establishes what a change of metric preserves, then states the additional conditions that identify a local loading class.

3 Metric-Indexed Identification

Two questions must be separated. Can the same fitted index be expressed under another metric? Once a metric is fixed, does the population criterion distinguish the chosen loading class from nearby alternatives? The first is a representation result; the second is local identification.

3.1 Metric sections and fitted-index equivalence

Stack the retained realizations and loadings as $v_{t+1} = (g'_{t+1}, 1)'$ and $\Gamma_v = [\Gamma_\delta, \Gamma_\alpha]$. Then $h_{t+1} = \Gamma_\beta f_{t+1} + \Gamma_v v_{t+1}$. In addition to a nonsingular change of latent basis, the transformation

$$f_{t+1}^{[D_v]} = f_{t+1} + D_v v_{t+1}, \quad \Gamma_v^{[D_v]} = \Gamma_v - \Gamma_\beta D_v \quad (3.1)$$

preserves the complete index for every $D_v \in \mathbb{R}^{K \times (K_o + 1)}$. Thus removing latent rotations alone does not determine the allocation between latent and retained blocks.

Suppose Γ_β has rank K . For a declared metric $M \succ 0$, set

$$R_M = (\Gamma'_\beta M \Gamma_\beta)^{1/2}, \quad \Gamma_\beta^M = \Gamma_\beta R_M^{-1}, \quad (3.2)$$

$$C_M = (\Gamma_\beta^M)' M \Gamma_v, \quad \Gamma_v^M = \Gamma_v - \Gamma_\beta^M C_M, \quad (3.3)$$

$$f_{t+1}^M = R_M f_{t+1} + C_M v_{t+1}. \quad (3.4)$$

Normalization changes the latent units; projection removes the retained loadings' M -overlap with that span. The factor adjustment offsets both operations.

Proposition 1 (Metric-section equivalence). *For every $M \succ 0$, the transformed representation satisfies*

$$(\Gamma_\beta^M)' M \Gamma_\beta^M = I_K, \quad (\Gamma_\beta^M)' M \Gamma_v^M = 0, \quad \Gamma_\beta^M f_{t+1}^M + \Gamma_v^M v_{t+1} = h_{t+1}. \quad (3.5)$$

Within the orbit generated by nonsingular latent-basis changes and (3.1), the resulting loading representative is unique up to orthogonal latent rotations. The result imposes the equality restrictions, not the exposure inequality.

Indeed, $R_M^2 = \Gamma'_\beta M \Gamma_\beta$ gives the first equality. The definition of C_M gives the second, and the two terms involving $\Gamma_\beta^M C_M v_{t+1}$ cancel in the complete index. To see uniqueness within the specified orbit, write another normalized latent block as $\Gamma_\beta^M G$. Its normalization implies $G'G = I_K$. Orthogonality of its retained block then forces the remaining cross-block reallocation to be zero. The proposition concerns this transformation orbit; it does not identify every representation capable of fitting a given finite panel.

Corollary 1 (Invariance of residual-based fit). *Any statistic depending on the representation only through its fitted indices is unchanged by (3.2)–(3.4), when the data, date weights, and included blocks are held fixed.*

This follows observation by observation from (3.5). In particular, residual sums of squares and R^2 cannot select among matched metric sections. The result gives an empirical comparison with fit held constant: changes in reported benchmark or intercept exposure can be attributed to the measurement convention. A finite exposure budget is different. Once a section is fixed, $\Gamma'_\alpha Q \Gamma_\alpha$ is constant over latent rotations, so the budget admits or excludes that class. Changing this feasible set can change the criterion target and fitted returns.

3.2 Local coordinates and identification

For a fixed metric, identification asks whether the population criterion separates nearby loading classes. A chart removes the rotational directions before making this comparison. Let $B_0 = \Gamma_{\beta,0}$, $A_0 = [\Gamma_{\delta,0}, \Gamma_{\alpha,0}]$, and choose $U_0 = Q_Z^{1/2} B_0$ and $U_{0\perp}$ so that $[U_0, U_{0\perp}]$ is orthogonal. For $C \in \mathbb{R}^{(L-K) \times K}$ and $D \in \mathbb{R}^{(L-K) \times (K_o+1)}$, define

$$U(C) = (U_0 + U_{0\perp} C)(I_K + C' C)^{-1/2}, \quad (3.6)$$

$$B(C; Q) = Q^{-1/2} U(C), \quad (3.7)$$

$$A(C, D; Q) = \{I_L - B(C; Q)B(C; Q)'Q\}\{A_0 + Q^{-1/2}U_{0\perp}D\}. \quad (3.8)$$

With $\eta = (\text{vec}(C)', \text{vec}(D)')'$, let $\chi_Q(\eta) = [B(C; Q), A(C, D; Q)]$. These maps are smooth and satisfy the equality restrictions. Locally, they are one-to-one on loading classes: if $Y = Q^{1/2}B$ is a nearby orthonormal basis, then $C = U'_{0\perp} Y (U'_0 Y)^{-1}$, independent of its right rotation. Given C , the retained coordinate is recovered using

$$D = \{U'_{0\perp}(I_L - UU')U_{0\perp}\}^{-1}U'_{0\perp}\{Q^{1/2}A - (I_L - UU')Q^{1/2}A_0\}.$$

Both inverse matrices are nonsingular near the origin, where the relevant overlap matrices are identities. This verifies the local chart and dimension (2.9); the construction in the Online Supplement gives further implementation details.

Let $\tilde{Q}(\eta; Q)$ be the population criterion expressed in this chart. Section 4 specifies the exactly profiled criterion used throughout. We treat gradients in η as columns.

Assumption 1 (Population local identification). *At $\chi_{Q_Z}(0_d) = [\vartheta_0]$, the population criterion is twice continuously differentiable and*

$$\Gamma'_{\alpha,0} Q_Z \Gamma_{\alpha,0} < \delta_\alpha, \quad S(0_d, Q_Z) = 0_d, \quad H := D_\eta S(0_d, Q_Z) \succeq c_H I_d \quad (3.9)$$

for $c_H > 0$, where $S = D_\eta \tilde{Q}$.

Slackness gives an open feasible neighborhood; stationarity removes first-order drift; curvature separates distinct classes.

Proposition 2 (Local quadratic identification). *Under Assumption 1, for sufficiently small η and some $c > 0$,*

$$\tilde{Q}(\eta; Q_Z) - \tilde{Q}(0_d; Q_Z) \geq c \|\eta\|^2. \quad (3.10)$$

Consequently, $[\vartheta_0]$ is the unique local minimizer on this chart.

Taylor expansion gives the criterion difference as $\frac{1}{2}\eta'H\eta + o(\|\eta\|^2)$. Shrinking the neighborhood makes the remainder at most $c_H\|\eta\|^2/4$ in absolute value, giving $c = c_H/4$. Chart injectivity converts separation of coordinates into separation of loading classes. This is local identification under the stated criterion.

The inequality gives statistical content to separation: moving to any distinct nearby class incurs a positive population loss. It therefore provides the gap needed for consistency. In economic applications the metric makes benchmark and intercept exposures comparable, while curvature determines whether the data distinguish those exposures locally. The next section constructs the estimator whose loss makes this distinction.

4 Fixed-Weight Profiled Estimation

4.1 The criterion and exact factor profile

Write $B = \Gamma_\beta$, $A = [\Gamma_\delta, \Gamma_\alpha]$, and $c_{t+1} = (g'_{t+1}, 1)'$. For candidate loadings and a date-specific factor, the candidate index is $\mathbf{h}_{t+1} = Bf + Ac_{t+1}$. Define the date loss and managed residual by

$$q_t(B, A, f) = \frac{1}{2}\mathbf{h}'_{t+1}W_t\mathbf{h}_{t+1} - X'_{t+1}\mathbf{h}_{t+1}, \quad m_t = X_{t+1} - W_t\mathbf{h}_{t+1}. \quad (4.1)$$

The loss is unit-level least squares up to a data-only constant:

$$\frac{\|r_{t+1} - Z_t\mathbf{h}_{t+1}\|^2}{2N_t} = q_t(B, A, f) + \frac{r'_{t+1}r_{t+1}}{2N_t}. \quad (4.2)$$

It also equals $\frac{1}{2}m'_tW_t^+m_t$ up to the constant $\frac{1}{2}X'_{t+1}W_t^+X_{t+1}$, where W_t^+ is the Moore–Penrose inverse. To verify this identity, use $X_{t+1} \in \text{Range}(W_t)$ and $W_tW_t^+W_t = W_t$. Thus the managed criterion preserves the least-squares problem even when W_t is singular; it does not impose an efficient weighting matrix.

Whenever $B'W_tB$ is positive definite, the factor normal equation $B'm_t = 0_K$ has the unique solution

$$f_{t+1}^*(B, A) = (B'W_tB)^{-1}B'(X_{t+1} - W_tAc_{t+1}). \quad (4.3)$$

Let $\tilde{q}_t(B, A) = q_t\{B, A, f_{t+1}^*(B, A)\}$ and $\tilde{m}_t(B, A) = m_t\{B, A, f_{t+1}^*(B, A)\}$. The envelope

theorem gives the profiled loading score in any ambient loading direction $(\dot{B}, \dot{A})$:

$$D\tilde{q}_t(B, A)[\dot{B}, \dot{A}] = -\tilde{m}'_t(\dot{B}f_{t+1}^* + \dot{A}c_{t+1}). \quad (4.4)$$

Here ambient means before imposing the metric restrictions and quotient. The derivative of the optimized factor drops out because $B'\tilde{m}_t = 0$. It must, however, be retained when differentiating this score. Differentiating the normal equation, with $G_t = (B'W_tB)^{-1}$, gives

$$Df_{t+1}^*[\dot{B}, \dot{A}] = G_t\{\dot{B}'\tilde{m}_t - B'W_t(\dot{B}f_{t+1}^* + \dot{A}c_{t+1})\}. \quad (4.5)$$

These two identities specify the exact profile derivatives used below. They link computation to inference: factors can be removed from the optimization variables without discarding their response from the loading Hessian. The remaining optimization is over loading classes, with the date-specific least-squares fit preserved.

4.2 The local estimator

Let $a_{t,T}$ be deterministic date weights with $T^{-1} \sum_t a_{t,T} = 1$ and $0 < a_- \leq a_{t,T} \leq a_+ < \infty$. Equal weighting sets $a_{t,T} = 1$. Define the ambient and quotient criteria by

$$\begin{aligned} \tilde{\mathcal{L}}_T(B, A) &= \frac{1}{T} \sum_{t=1}^T a_{t,T} \tilde{q}_t(B, A), \\ \tilde{Q}_T(\eta; Q) &= \tilde{\mathcal{L}}_T\{B(\eta; Q), A(\eta; Q)\}, \quad \tilde{Q}(\eta; Q) = \mathbb{E}\tilde{Q}_T(\eta; Q). \end{aligned} \quad (4.6)$$

The three weighting objects have distinct roles: W_t^+ reproduces within-date least squares, $a_{t,T}$ determines each date's contribution, and Q selects the loading representation. The date weights do not redefine the sample metric $\hat{Q}_Z = T^{-1} \sum_t W_t$. The term fixed-weight refers to the deterministic date array, not to a known metric.

On a sufficiently small compact coordinate neighborhood $\overline{\mathcal{U}}_0$ containing 0_d in its interior, define

$$\hat{\eta}_T \in \arg \min_{\eta \in \overline{\mathcal{U}}_0} \tilde{Q}_T(\eta; \hat{Q}_Z), \quad [\hat{\vartheta}_T] = \chi_{\hat{Q}_Z}(\hat{\eta}_T). \quad (4.7)$$

For the interior theory, choose the neighborhood so that its chart representatives satisfy the exposure budget for all nearby metrics. This is a local estimator on a declared branch. An algorithm implements it only if it selects that branch and solves the profiled problem

to the required accuracy. A finite number of alternating updates can provide an initializer but does not by itself establish this property. Before analyzing its sampling error, we ask what population object that local fit estimates. This is the step that connects a well-solved numerical criterion to a structural interpretation of the loadings.

4.3 Structural and criterion targets

The criterion target $[\vartheta_0]$ minimizes expected profiled loss. The structural class $[\vartheta^\circ]$ instead specifies the loadings in the data-generating factor equation; the superscript $\circ$ marks this distinction. Whether these classes coincide determines whether inference describes structural exposures or the best local approximation selected by the criterion. With a finite cross section, profiling can make the distinction consequential.

Let (B°, A°) be the structural representative at Q_Z and write $X_{t+1} = W_t(B^\circ f_{t+1}^\circ + A^\circ c_{t+1}) + u_{t+1}$. Set

$$G_t^\circ = (B^{\circ\prime} W_t B^\circ)^{-1}, \quad R_t^\circ = I_L - W_t B^\circ G_t^\circ B^{\circ\prime}. \quad (4.8)$$

Substitution into (4.3) yields

$$f_{t+1}^*(B^\circ, A^\circ) = f_{t+1}^\circ + G_t^\circ B^{\circ\prime} u_{t+1}, \quad \tilde{m}_t(B^\circ, A^\circ) = R_t^\circ u_{t+1}. \quad (4.9)$$

The fitted factor absorbs part of the disturbance while the rest remains in the residual. The loading score multiplies these two objects, producing a quadratic disturbance term even under conditional mean zero. For example, fitting a factor to a finite set of stocks uses some of their idiosyncratic return variation. That same variation also enters the score used to choose the loading span. Score centering, rather than disappearance of every fitted-factor error, is what the structural interpretation needs.

Let $\mathcal{C}_t$ be a sigma-field with respect to which $Z_t, c_{t+1}, f_{t+1}^\circ$ are measurable.

Assumption 2 (Design-aligned managed disturbance). *For a nonnegative $\mathcal{C}_t$ -measurable scalar ω_t ,*

$$\mathbb{E}(u_{t+1} \mid \mathcal{C}_t) = 0_L, \quad \mathbb{E}(u_{t+1} u_{t+1}' \mid \mathcal{C}_t) = \omega_t W_t. \quad (4.10)$$

Conditional unit-level sphericity is sufficient: if $\mathbb{E}(\varepsilon_{t+1} \mid \mathcal{C}_t) = 0$ and $\text{Var}(\varepsilon_{t+1} \mid \mathcal{C}_t) = \sigma_t^2 I_{N_t}$, then $\omega_t = \sigma_t^2 / N_t$. Neither Gaussianity nor independence across units is needed for this implication.

Proposition 3 (Structural interpretation of the profiled target). *Suppose the factor profile is regular at (B°, A°) , local loss derivatives admit integrable envelopes permitting differentiation under expectation, and Assumption 2 holds. The population quotient score then vanishes at the structural class. If the budget is strictly slack and the quotient Hessian is positive definite there, $[\vartheta_0] = [\vartheta^\circ]$ on the selected local branch.*

To establish score centering, substitute (4.9) into (4.4). Terms linear in u_{t+1} have zero conditional expectation. The quadratic term, using $x'Kx = \text{tr}(Kxx')$, gives

$$\mathbb{E}\{D\tilde{q}_t(B^\circ, A^\circ)[\dot{B}, \dot{A}] \mid \mathcal{C}_t\} = -\text{tr}\{R_t^{\circ'} \dot{B} G_t^\circ B^{\circ'} \mathbb{E}(u_{t+1} u_{t+1}' \mid \mathcal{C}_t)\}. \quad (4.11)$$

Covariance alignment makes this $-\omega_t \text{tr}\{G_t^\circ B^{\circ'} W_t R_t^{\circ'} \dot{B}\} = 0$, because $R_t^\circ W_t B^\circ = 0$. Deterministic aggregation and the chain rule give a zero quotient score. Proposition 2 then establishes the local minimum.

The condition removes profile-induced score bias without requiring each date's factor error to vanish. It therefore permits structural inference with fixed N_t and increasing T . Covariance alignment is sufficient, not necessary: the same interpretation holds whenever the population quotient score is centered at the structural class and local separation holds. Otherwise the inference below concerns the criterion target. This distinction is separate from a joint large- N, T approximation. In practice, the result tells the researcher which additional disturbance restriction makes an interval for a fitted loading comparison an interval for the structural comparison. Consistency for the criterion target alone does not answer that interpretive question.

4.4 Regularity and consistency

Write $S_T = D_\eta \tilde{Q}_T$ and $S = D_\eta \tilde{Q}$, with gradients interpreted as columns. The following conditions make the local derivatives and localization arguments precise.

Assumption 3 (Sampling and metric estimation). *Define the date vector*

$$Y_t = (X_{t+1}', \text{vec}(W_t)', c_{t+1}')'.$$

It is strictly stationary and alpha mixing, with $\sum_{j \geq 1} \alpha(j)^{\delta/(2+\delta)} < \infty$ and $\mathbb{E}\|Y_t\|^{4+\delta} < \infty$ for some $\delta > 0$. The date weights satisfy the deterministic bounds and normalization above, and $Q_Z = \mathbb{E}W_t \succ 0$.

Stationarity and weight normalization make the expected criterion independent of T . The sample Gram metric has the exact representation

$$\sqrt{T} \operatorname{vec}(\widehat{Q}_Z - Q_Z) = \frac{1}{\sqrt{T}} \sum_{t=1}^T \varphi_{Q,t}, \quad \varphi_{Q,t} = \operatorname{vec}(W_t - Q_Z). \quad (4.12)$$

The mixing covariance inequality and the stated moments give a finite partial-sum variance, so $\widehat{Q}_Z - Q_Z = O_p(T^{-1/2})$. In particular, $\widehat{Q}_Z$ is positive definite with probability tending to one.

Assumption 4 (Uniform local regularity). *Assumption 1 holds. There are a compact convex $\overline{\mathcal{U}}_0$, a compact metric neighborhood $\mathcal{N}_Q \subset \mathbb{S}_{++}^L$, and a loading neighborhood $\mathcal{N}_B$ containing all $B(\eta; Q)$ on these sets, such that:*

- (i) *The representative map $(\eta, Q) \mapsto (B, A)$ is C^2 on an open neighborhood of the sets, and χ_Q is a local diffeomorphism onto its image. All these representatives satisfy the strictly slack budget.*
- (ii) *For some $c_M > 0$,*

$$\Pr \left\{ \inf_{1 \leq t \leq T} \inf_{B \in \mathcal{N}_B} \lambda_{\min}(B' W_t B) \geq c_M \right\} \longrightarrow 1. \quad (4.13)$$

- (iii) *For $\psi_{t,T}(\eta, Q) = a_{t,T} \widetilde{q}_t \{B(\eta; Q), A(\eta; Q)\}$, each of $\psi_{t,T}$, $D_\eta \psi_{t,T}$, $D_\eta^2 \psi_{t,T}$, $D_Q D_\eta \psi_{t,T}$ is measurable and continuous. Its norm is bounded by M_t , and its increment between two arguments is bounded by $L_t(\|\eta - \eta'\| + \|Q - Q'\|_F)$, uniformly in t, T . The nonnegative stationary envelopes satisfy $\mathbb{E}(M_t^{2+\delta} + L_t^{2+\delta}) < \infty$.*

These conditions extend local identification to a neighborhood in which feasible estimation can be analyzed. Part (i) keeps nearby loading classes on a smooth, strictly feasible branch as the metric varies. Part (ii) requires the characteristics to support the K latent directions uniformly across dates and nearby loadings. It does not require the entire Gram matrix W_t to have full rank. The uniform lower bound keeps $(B' W_t B)^{-1}$ bounded with probability approaching one, preventing ill-conditioned factor solves from destabilizing the profiled criterion. This is stronger than requiring a unique factor solution at each date. Part (iii) controls the size and local variation of the loss and its derivatives on the same neighborhood. Together with the sampling conditions, these bounds permit uniform sample approximation, including evaluation at the estimated metric. The following proposition uses this approximation to translate population separation into consistency of the selected loading

class.

Proposition 4 (Local consistency). *Under Assumptions 3 and 4, $\hat{\eta}_T \rightarrow_p 0_d$ and the estimated class converges locally to $[\vartheta_0]$. The same conclusion holds for a selected-branch estimator with an $o_p(1)$ objective gap.*

For completeness, a finite covering of the compact parameter sets reduces uniform convergence to convergence at finitely many arguments. At each argument the mixing covariance inequality bounds the variance of the centered average by a constant times T^{-1} . The Lipschitz envelope bounds the error between a point and its covering point by its distance times $T^{-1} \sum_t L_t = O_p(1)$. Letting the covering radius decrease proves the uniform law for the criterion and its displayed derivatives. Continuity in Q then permits substitution of $\hat{Q}_Z$. For any fixed radius $\varepsilon > 0$, quadratic separation gives a positive criterion gap outside that radius within a sufficiently small $\overline{\mathcal{U}}_0$. Uniform approximation and the vanishing optimization gap prevent a minimizer from remaining there. This proves consistency, including localization away from the artificial chart boundary.

Consistency places the estimated class near the population target, so the local derivatives can be used at the estimate. It does not yet measure sampling precision. For that purpose, we next identify how learning the metric moves the target and then derive its first-order sampling law.

4.5 Metric sensitivity

Hold the sampling law fixed. For the population quotient score S , let $H = D_\eta S(0_d, Q_Z) \in \mathbb{R}^{d \times d}$ denote the positive-definite quotient Hessian. Define $G_Q \in \mathbb{R}^{d \times L^2}$ by

$$D_Q S(0_d, Q_Z)[\Delta Q] = G_Q \text{vec}(\Delta Q), \quad \Delta Q \in \mathbb{S}^L, \quad (4.14)$$

where $\mathbb{S}^L$ is the space of symmetric matrices and vec stacks columns. This derivative accounts jointly for the metric-dependent representative and the exact factor profile. Its matrix representation need only agree on symmetric perturbations.

Let $\eta_0(Q)$ denote the local population stationary branch satisfying $S\{\eta_0(Q), Q\} = 0_d$ and $\eta_0(Q_Z) = 0_d$. The implicit function theorem gives

$$D_Q \eta_0(Q_Z)[\Delta Q] = -H^{-1} G_Q \text{vec}(\Delta Q). \quad (4.15)$$

The metric perturbation changes the score by $G_Q \text{vec}(\Delta Q)$; the inverse Hessian converts

it into the coordinate adjustment that restores stationarity. For the estimated Gram metric, recall from (4.12) that $\varphi_{Q,t} = \text{vec}(W_t - Q_Z) \in \mathbb{R}^{L^2}$ is its date-level influence contribution. Thus $G_Q \varphi_{Q,t}$ is the corresponding score contribution used in Section 5.

Let $\phi(\eta, Q) \in \mathbb{R}^{p_\phi}$ be the local expression of a prespecified smooth rotation-invariant report, where p_ϕ is the number of reported statistics; Section 6 develops its inference. Define $\phi_\eta = D_\eta \phi(0_d, Q_Z) \in \mathbb{R}^{p_\phi \times d}$ and $\phi_Q \in \mathbb{R}^{p_\phi \times L^2}$ by $D_Q \phi(0_d, Q_Z)[\Delta Q] = \phi_Q \text{vec}(\Delta Q)$. The chain rule yields

$$D_Q[\phi\{\eta_0(Q), Q\}]_{Q_Z}[\Delta Q] = (\phi_Q - \phi_\eta H^{-1} G_Q) \text{vec}(\Delta Q). \quad (4.16)$$

The two terms capture direct metric dependence and movement of the target. Along the matched section path on this interior branch, they cancel for smooth functions of the raw latent span and fixed benchmarks. They need not cancel for the metric-whitened projector, whose definition itself changes with the metric.

Prespecified smooth metric transformations fit the same construction. For a fixed C^1 map g into the positive-definite cone, replace $\widehat{Q}_Z$ and Q_Z by $g(\widehat{Q}_Z)$ and $g(Q_Z)$, recenter the chart at the corresponding target, and reevaluate the criterion and report derivatives. Provided the transformed specification satisfies the maintained local regularity conditions, the delta method gives the metric influence

$$\varphi_{Q,t}^g = \text{vec}\{Dg_{Q_Z}[W_t - Q_Z]\}. \quad (4.17)$$

This covers trace normalization and fixed smooth shrinkage. Data-dependent tuning or nonsmooth repairs require additional analysis.

5 Local Asymptotic Theory under a Learned Metric

The same observations determine the profiled criterion and the metric. The feasible score must combine both channels before a covariance is formed. Define the date contributions at the population target by

$$s_{t,T}^{\text{prof}} = D_\eta \psi_{t,T}(0_d, Q_Z), \quad \xi_{t,T} = s_{t,T}^{\text{prof}} + G_Q \varphi_{Q,t}. \quad (5.1)$$

We suppress the second subscript when no ambiguity arises. Stationarity, deterministic weights, and $S(0_d, Q_Z) = 0$ imply $\mathbb{E}s_{t,T}^{\text{prof}} = 0$ for every date. The complete score $\xi_{t,T}$ retains

both profiled return-score variation and the change in the first-order condition caused by learning the metric. This is the two-step correction for the present metric-indexed estimand (Newey, 1994; Newey and McFadden, 1994).

Theorem 1 (Interior asymptotic linear representation). *Under Assumptions 3 and 4,*

$$\sqrt{T} \hat{\eta}_T = -H^{-1} \frac{1}{\sqrt{T}} \sum_{t=1}^T \xi_{t,T} + o_p(1). \quad (5.2)$$

The same representation holds for a consistent selected-branch numerical solution $\tilde{\eta}_T$ satisfying

$$\|S_T(\tilde{\eta}_T, \hat{Q}_Z)\| = o_p(T^{-1/2}). \quad (5.3)$$

Proof. The mixing covariance inequality and the $2 + \delta$ envelopes bound the variance of the normalized partial sums of $(s_{t,T}^{\text{prof}}, \varphi'_{Q,t})'$ uniformly in T . Their means are zero, so these sums are $O_p(1)$. Apply the integral Taylor formula in Q to $S_T(0_d, Q)$. The uniform derivative law established in the proof of Proposition 4, continuity of its expectation, and $\hat{Q}_Z - Q_Z = O_p(T^{-1/2})$ give

$$\sqrt{T} S_T(0_d, \hat{Q}_Z) = \frac{1}{\sqrt{T}} \sum_{t=1}^T \xi_{t,T} + o_p(1). \quad (5.4)$$

Explicitly, the remainder is $\sqrt{T}$ times the metric displacement multiplied by the difference between the integrated sample metric derivative and G_Q ; that difference is $o_p(1)$.

Consistency and strict slackness put the exact minimizer in the chart interior with probability tending to one. Its score is then zero. A second integral expansion gives

$$0 = S_T(0_d, \hat{Q}_Z) + \bar{H}_T \hat{\eta}_T, \quad \bar{H}_T = \int_0^1 D_\eta S_T(v \hat{\eta}_T, \hat{Q}_Z) dv.$$

Uniform derivative convergence and consistency imply $\bar{H}_T \rightarrow_p H \succ 0$. Inversion first gives $\hat{\eta}_T = O_p(T^{-1/2})$ and then (5.2). Replacing the zero score by the residual in (5.3) adds only $o_p(1)$ after root- T scaling, proving the numerical statement. $\square$

The theorem explains how sampling uncertainty moves the estimated decomposition: the complete score supplies the disturbance and the inverse quotient Hessian determines its amplification. An $o_p(1)$ objective gap supports consistency but need not make numerical error negligible at this scale. Condition (5.3) separates these two requirements.

For a unit eigenvector w_j of H with eigenvalue $\lambda_j > 0$,

$$w_j' \sqrt{T} \hat{\eta}_T = -\lambda_j^{-1} w_j' \frac{1}{\sqrt{T}} \sum_t \xi_{t,T} + o_p(1). \quad (5.5)$$

Within the selected chart, w_j describes a direction of change in the loading class. Lower curvature amplifies a fixed projected disturbance. A locally unique decomposition can therefore be measured imprecisely, even when fitted returns are stable. Inference for empirical exposures uses report derivatives rather than interpreting these coordinate eigenvectors as economic factors.

An asymptotic linear representation alone does not specify normal critical values. The next condition supplies the joint distribution of its input, including the metric shock that a report may also use directly.

Assumption 5 (Joint complete-score and metric limit). *For $y_{t,T} = (\xi'_{t,T}, \varphi'_{Q,t})'$ and a finite positive-semidefinite matrix $\mathcal{J}_{\xi Q}$,*

$$\text{Var} \left(T^{-1/2} \sum_t y_{t,T} \right) \rightarrow \mathcal{J}_{\xi Q}, \quad T^{-1/2} \sum_t y_{t,T} \Rightarrow \mathcal{N}(0_{d+L^2}, \mathcal{J}_{\xi Q}). \quad (5.6)$$

A joint CLT is needed because a report or boundary may also respond directly to the metric. For equal weights the maintained mixing conditions provide the usual stationary CLT route. For a general deterministic weight array, covariance convergence is an additional stability condition, not a consequence of bounded weights alone.

Corollary 2 (Quotient sandwich law). *Under Theorem 1 and Assumption 5,*

$$\sqrt{T} \hat{\eta}_T \Rightarrow \mathcal{N}(0_d, V_\eta), \quad V_\eta = H^{-1} \mathcal{J}_\xi H^{-\top}, \quad (5.7)$$

where $\mathcal{J}_\xi$ is the complete-score block of $\mathcal{J}_{\xi Q}$.

The result follows by applying Slutsky's theorem to (5.2). If the component covariance limits exist, applying the covariance formula for a linear combination to the normalized partial sums of $\xi_t = s_{t,0}^{\text{prof}} + G_Q \varphi_{Q,t}$ and taking their long-run covariance limits gives

$$\mathcal{J}_\xi = \mathcal{J}_s + G_Q \mathcal{J}_Q G_Q' + \mathcal{J}_{sQ} G_Q' + G_Q \mathcal{J}_{sQ}'. \quad (5.8)$$

Here $\mathcal{J}_s$ and $\mathcal{J}_Q$ are the long-run covariances of the profiled score and metric influence, and

$\mathcal{J}_{sQ}$ is their long-run cross-covariance. The last two terms capture reinforcement or offsetting between the two channels. We do not impose $\mathcal{J}_{sQ} = 0$. Because the same characteristics enter the return criterion and the metric, adding their marginal variances is generally not the same as estimating the variance of their sum. Retaining the joint contribution permits either reinforcement or offsetting without having to choose between them in advance.

For example, the directional limiting variance is $w'_j \mathcal{J}_\xi w_j / \lambda_j^2$. Uncertainty can be large because the complete score is volatile, because curvature is small, or both. Section 6 maps this coordinate law into uncertainty about prespecified features of the loading class.

6 Invariant Reports and Feasible Covariance

Quotient coordinates permit differentiation, but empirical questions concern features of the loading class. A researcher may ask how closely the latent span aligns with a benchmark direction, how it differs from a benchmark span, or how much intercept exposure remains. This section attaches standard errors to those comparisons while retaining metric uncertainty.

6.1 Reports and empirical questions

For a declared metric Q , a report is a prespecified map

$$\Phi_Q : \mathcal{M}(Q, \delta_\alpha) / \mathcal{O}(K) \longrightarrow \mathbb{R}^{p_\phi}, \quad (6.1)$$

where p_ϕ is fixed. Its value is unchanged when B is replaced by BR , $R \in \mathcal{O}(K)$, with the retained loading blocks held fixed. This invariance makes the report a feature of the loading class rather than of a particular latent basis. A change of chart likewise changes its expression, not its value; the declared metric may remain part of the empirical comparison.

Write $\phi(\eta, Q) = \Phi_Q\{\chi_Q(\eta)\}$. The following assumption collects the report-specific conditions. Part (i) is used for the first-order expansion; part (ii) is required only for a normal interval.

Assumption 6 (Report regularity). *(i) The report has an extension $\phi \in C^1(\mathcal{V}; \mathbb{R}^{p_\phi})$, where $\mathcal{V}$ is an open neighborhood of $(0_d, Q_Z)$ in $\mathbb{R}^d \times \mathbb{S}_{++}^L$. At this point, write*

$$\phi_\eta = D_\eta \phi(0_d, Q_Z), \quad D_Q \phi(0_d, Q_Z)[\Delta Q] = \phi_Q \text{vec}(\Delta Q),$$

with derivative matrices of dimensions $p_\phi \times d$ and $p_\phi \times L^2$.

(ii) For the prespecified nonzero contrast $r_\phi \in \mathbb{R}^{p_\phi}$ used for normal inference,

$$\mathcal{A}_\phi = [-\phi_\eta H^{-1}, \phi_Q], \quad v_{r_\phi} := r'_\phi \mathcal{A}_\phi \mathcal{J}_{\xi Q} \mathcal{A}'_\phi r_\phi > 0.$$

Here H and $\mathcal{J}_{\xi Q}$ are the quotient Hessian and joint score–metric long-run covariance from Section 5. Part (i) permits differentiation jointly in loadings and metric; the extension also defines derivatives at a boundary target. Part (ii) requires the chosen comparison to retain first-order sampling variation. It does not require the full report covariance to be positive definite. The next subsection derives this covariance, and Assumption 7 states the additional conditions for estimating it. Benchmarks are fixed here; an estimated benchmark requires its sampling contribution in the joint expansion.

For a fixed nonzero direction $a_\star \in \mathbb{R}^L$, a fixed full-column-rank benchmark matrix $D_\star$ with L rows, and the whitened-span projector $\mathcal{P}_Q$ in (2.10), three scalar examples are

$$\Phi_{a_\star, Q}([\vartheta]) = \frac{a'_\star Q B (B' Q B)^{-1} B' Q a_\star}{a'_\star Q a_\star}, \quad (6.2)$$

$$\Phi_{D_\star, Q}([\vartheta]) = \|\mathcal{P}_Q(B) - \mathcal{P}_Q(D_\star)\|_F^2, \quad (6.3)$$

$$\Phi_{\alpha, Q}([\vartheta]) = \Gamma'_\alpha Q \Gamma_\alpha. \quad (6.4)$$

They measure alignment with a characteristic direction, distance from a benchmark span, and retained intercept exposure. Positive definiteness of Q and the stated rank conditions keep the denominator positive and the Gram inverses well defined. Together with the smooth representative map, they establish part (i). These basis-invariant reports can be stacked to study the precision of comparisons across exposure measures.

Part (ii) needs a separate check even for these smooth examples. For exposure, the product rule gives

$$D(\Gamma'_\alpha Q \Gamma_\alpha)[\dot{\Gamma}_\alpha, \Delta Q] = 2\Gamma'_\alpha Q \dot{\Gamma}_\alpha + \Gamma'_\alpha \Delta Q \Gamma_\alpha.$$

Both terms vanish at $\Gamma_{\alpha,0} = 0$. With a positive slack budget, the loading estimator can still have root- T error, but its squared exposure is then only $O_p(T^{-1})$. Here a zero first-order variance reflects a locally flat report, not an exactly known loading class. The same mechanism applies to squared alignment at exact alignment or orthogonality and to squared span distance at equal spans. These comparisons require higher-order analysis or a different nondegenerate report, rather than the normal interval below.

Researchers can define other reports using the same two checks: smoothness governs the expansion, and the chosen contrast's variance governs normal inference. A binding exposure budget raises a distinct issue by changing the loading estimator's local law. Section 7 treats that case under its additional uniform approximation and critical-quantile conditions.

6.2 Influence and chart invariance

Let $\widehat{\phi}_T = \phi(\widehat{\eta}_T, \widehat{Q}_Z)$ and $\phi_0 = \phi(0_d, Q_Z)$. Use the derivative matrices defined in Assumption 6(i).

Theorem 2 (Invariant reports). *Under Theorem 1 and Assumption 6(i),*

$$\sqrt{T}(\widehat{\phi}_T - \phi_0) = \frac{1}{\sqrt{T}} \sum_{t=1}^T \zeta_{\phi,t,T} + o_p(1), \quad (6.5)$$

$$\zeta_{\phi,t,T} = -\phi_\eta H^{-1} \xi_{t,T} + \phi_Q \varphi_{Q,t}. \quad (6.6)$$

This influence contribution is unchanged by an admissible smooth change of quotient chart, including a metric-dependent change. Under Assumption 5, the limit is Gaussian with covariance

$$V_\phi = \mathcal{A}_\phi \mathcal{J}_{\xi Q} \mathcal{A}'_\phi. \quad (6.7)$$

Proof. Write $\widehat{\phi}_T = \phi(\widehat{\eta}_T, \widehat{Q}_Z)$ and $\phi_0 = \phi(0_d, Q_Z)$. The derivative matrices ϕ_η and ϕ_Q are evaluated at $(0_d, Q_Z)$ and describe the report's responses to the quotient coordinate and metric, respectively. The joint delta method gives

$$\sqrt{T}(\widehat{\phi}_T - \phi_0) = \phi_\eta \sqrt{T} \widehat{\eta}_T + \phi_Q \sqrt{T} \text{vec}(\widehat{Q}_Z - Q_Z) + o_p(1).$$

Let ξ_t denote the complete-score contribution and $\varphi_{Q,t}$ the metric-estimation influence contribution in (5.2) and (4.12). Substituting those representations gives the report influence

$$\zeta_{\phi,t} = -\phi_\eta H^{-1} \xi_t + \phi_Q \varphi_{Q,t} = \mathcal{A}_\phi \begin{pmatrix} \xi_t \\ \varphi_{Q,t} \end{pmatrix}, \quad \mathcal{A}_\phi = [-\phi_\eta H^{-1}, \phi_Q],$$

where H is the population quotient Hessian. Applying this linear map to the joint CLT gives covariance $V_\phi = \mathcal{A}_\phi \mathcal{J}_{\xi Q} \mathcal{A}'_\phi$, where $\mathcal{J}_{\xi Q}$ is the long-run covariance of the stacked score and metric influences.

For chart invariance, let $\bar{\eta} = \tau_Q(\eta)$ be a smooth, locally invertible transition to another coordinate system for the same loading class. Both reference coordinates are zero at $Q = Q_Z$. Define its coordinate and metric derivatives by

$$J = D_{\eta}\tau_{Q_Z}(0_d) \in \mathbb{R}^{d \times d}, \quad D_Q\tau_{Q_Z}(0_d)[\Delta Q] = K_Q \text{vec}(\Delta Q), \quad K_Q \in \mathbb{R}^{d \times L^2},$$

where ΔQ is a symmetric metric perturbation and J is nonsingular. Let $\ell_t = -H^{-1}\xi_t$ be the influence contribution for $\hat{\eta}_T$. Applying the chain rule to $\hat{\eta}_T = \tau_{\hat{Q}_Z}(\hat{\eta}_T)$ gives its counterpart $\bar{\ell}_t = J\ell_t + K_Q\varphi_{Q,t}$.

Let $\bar{\phi}(\bar{\eta}, Q)$ express the same report in the new chart, so that $\phi(\eta, Q) = \bar{\phi}\{\tau_Q(\eta), Q\}$. Define $\bar{\phi}_{\eta} = D_{\bar{\eta}}\bar{\phi}(0_d, Q_Z)$ and $D_Q\bar{\phi}(0_d, Q_Z)[\Delta Q] = \bar{\phi}_Q \text{vec}(\Delta Q)$. The chain rule then yields

$$\phi_{\eta} = \bar{\phi}_{\eta}J, \quad \phi_Q = \bar{\phi}_{\eta}K_Q + \bar{\phi}_Q.$$

Consequently, the report influence computed in the new chart satisfies

$$\begin{aligned} \bar{\phi}_{\eta}\bar{\ell}_t + \bar{\phi}_Q\varphi_{Q,t} &= \bar{\phi}_{\eta}J\ell_t + (\bar{\phi}_{\eta}K_Q + \bar{\phi}_Q)\varphi_{Q,t} \\ &= \phi_{\eta}\ell_t + \phi_Q\varphi_{Q,t} = \zeta_{\phi,t}. \end{aligned}$$

The complete report influence is therefore identical date by date in the two charts, as is its long-run covariance. $\square$

Equivalently, the metric coefficient in the report influence is $\phi_Q - \phi_{\eta}H^{-1}G_Q$, the population total derivative in (4.16). It is zero for a metric-invariant matched-span comparison, but generally not for a metric-whitened comparison. Keeping both terms ensures that an arbitrary coordinate change creates neither spurious uncertainty nor an omitted metric contribution. The practical consequence is that researchers can choose a numerically convenient local chart and still report the same standard errors for the same empirical question. The invariance concerns computation; a report whose definition changes with the metric may still measure a different exposure comparison under another metric.

6.3 Feasible long-run covariance

The loading estimate and report have already been constructed. What remains is to estimate the report's uncertainty from its complete influence sequence, preserving both within-date dependence and dependence across dates.

Evaluate the exactly profiled date scores in one sample reference chart, and estimate $H, G_Q, \phi_\eta, \phi_Q$ in that same convention. For the sample-average metric, construct

$$\begin{aligned}\widehat{\varphi}_{Q,t} &= \text{vec}(W_t - \widehat{Q}_Z), & \widehat{\xi}_t &= \widehat{s}_t^{\text{prof}} + \widehat{G}_Q \widehat{\varphi}_{Q,t}, \\ \widehat{\zeta}_{\phi,t} &= -\widehat{\phi}_\eta \widehat{H}^{-1} \widehat{\xi}_t + \widehat{\phi}_Q \widehat{\varphi}_{Q,t}.\end{aligned}\tag{6.8}$$

For a smooth metric transform use its transformed influence. The datewise construction retains the score–metric cross-covariances before any temporal covariance is estimated.

Center the estimated sequence as $\widetilde{\zeta}_t = \widehat{\zeta}_{\phi,t} - T^{-1} \sum_j \widehat{\zeta}_{\phi,j}$. For $0 \leq \ell \leq b_T$ let $\widehat{C}_\ell = T^{-1} \sum_{t=\ell+1}^T \widetilde{\zeta}_t \widetilde{\zeta}_{t-\ell}'$. The kernel HAC estimator is

$$\widehat{V}_\phi = \widehat{C}_0 + \sum_{\ell=1}^{b_T} k(\ell/b_T) (\widehat{C}_\ell + \widehat{C}_\ell').\tag{6.9}$$

Bartlett weights give a positive-semidefinite construction (Newey and West, 1987; Andrews, 1991). The following explicit conditions also cover the weighted influence array.

Assumption 7 (HAC approximation). *Write $\zeta_t = \zeta_{\phi,t,T}$ and define*

$$C_{\ell,T} = T^{-1} \sum_{t=\ell+1}^T \mathbb{E}(\zeta_t \zeta_{t-\ell}').$$

Define $C_{\ell,T}^o$ by replacing expectation with the observed product. For constants independent of T, ℓ :

- (i) $\mathbb{E}\zeta_t = 0$, $\sup_{t,T} \mathbb{E}\|\zeta_t\|^2 < \infty$, $\sup_T \sum_{\ell=0}^{T-1} \|C_{\ell,T}\| < \infty$, and $\lim_{m \rightarrow \infty} \sup_T \sum_{\ell=m}^{T-1} \|C_{\ell,T}\| = 0$. *The complete covariance sum converges to V_ϕ .*
- (ii) $\sup_{0 \leq \ell < T} \mathbb{E}\|C_{\ell,T}^o - C_{\ell,T}\|_F^2 \leq C/T$.
- (iii) *The kernel is bounded, symmetric, supported on $[-1, 1]$, continuous at zero, and $k(0) = 1$. The bandwidth satisfies $b_T \rightarrow \infty$ and $b_T/\sqrt{T} \rightarrow 0$. The estimated influences obey*

$$b_T \rho_T = o_p(1), \quad \rho_T = \left\{ T^{-1} \sum_t \|\widehat{\zeta}_{\phi,t} - \zeta_{\phi,t,T}\|^2 \right\}^{1/2}.\tag{6.10}$$

Parts (i)–(ii) state the required covariance-tail and lag-estimation bounds explicitly. Summable covariances and fourth-order cumulants provide familiar stationary sufficient conditions; for general date arrays the bounds must hold for the actual weighted array. They are additional covariance-estimation conditions, not conclusions from the CLT alone. Part

(iii) links derivative and score accuracy to the number of lags used.

Proposition 5 (Feasible report inference). *Under Theorem 2, Assumption 5, and Assumption 7, $\widehat{V}_\phi \rightarrow_p V_\phi$. If Assumption 6(ii) also holds, an asymptotic $(1 - \alpha)$ interval for $r'_\phi \phi_0$ is*

$$r'_\phi \widehat{\phi}_T \pm z_{1-\alpha/2} \sqrt{r'_\phi \widehat{V}_\phi r_\phi / T}, \quad 0 < \alpha < 1. \quad (6.11)$$

Here $z_{1-\alpha/2}$ is the $(1 - \alpha/2)$ quantile of the standard normal distribution, and α is the nominal noncoverage probability.

Proof. First use the true, uncentered influences. The lag-estimation bound and the triangle inequality make the stochastic error $O_p(b_T/\sqrt{T}) = o_p(1)$. For the expectation, split the covariance sum at a fixed lag m . Continuity of k handles the finitely many short lags, and part (i) bounds the tail uniformly; then let m increase. The joint CLT gives $T^{-1} \sum_t \zeta_t = O_p(T^{-1/2})$. Cauchy–Schwarz and the bounded average second moment therefore bound the centering error over the retained lags by $O_p(b_T/\sqrt{T})$. Finally, replacing true influences by estimated ones changes each lag product by at most a constant times $\rho_T(T^{-1} \sum_t \|\zeta_t\|^2)^{1/2} + \rho_T^2$. Centering does not increase the root-mean-square replacement error. Summing over lags and using (6.10) gives $o_p(1)$. Slutsky’s theorem proves the interval. $\square$

The proposition completes the interior inference procedure: estimate the loading class, evaluate the chosen report, construct its datewise influence, and estimate the long-run covariance of that sequence. In Section 8, Table II and Figure 2 show what changes when the metric part of this procedure is omitted while the point estimate is held fixed.

The proposition separates covariance estimation from interval construction: the covariance estimate can remain consistent when V_ϕ is singular, while a normal interval requires part (ii) for the chosen contrast. The next section studies how a binding exposure restriction changes the loading movement and the critical values.

7 A Learned Exposure Boundary

An exposure budget close to binding changes the shape, not just the variance, of the estimator’s sampling law. Metric estimation moves both the score and the boundary. We first describe their joint local experiment, then construct a feasible interval that does not require estimating the local slack.

7.1 Local drift and the uniform experiment

Let $P \in \mathfrak{P}_T$ denote one admissible data-generating law at sample size T ; a selection $P_T \in \mathfrak{P}_T$ is a triangular sequence. The factor equation defines a class of laws, not a unique distribution of characteristics, factors, and disturbances. Allowing population quantities to vary locally with T describes the transition between interior and binding inference.

For each law, center a chart at the equality-constrained, unrestricted population minimizer. Thus $\eta = 0_d$ solves the population score equation before the exposure inequality is imposed. Let

$$\mathfrak{c}_{\alpha,T}(\eta, Q) = \Gamma_{\alpha,T}(\eta; Q)' Q \Gamma_{\alpha,T}(\eta; Q) - \delta_{\alpha,T}, \quad \kappa_T = \sqrt{T} \mathfrak{c}_{\alpha,T}(0_d, Q_{Z,T}) \leq 0. \quad (7.1)$$

We suppress P in population quantities. Nonpositive excess means feasibility. If the population slack is $\rho/\sqrt{T}$, then $\kappa_T = -\rho$: the distance to the boundary is on the same scale as estimation error. Fixed positive slack gives $\kappa_T \rightarrow -\infty$; $\kappa_T \rightarrow 0$ includes exact binding and smaller-order slack. Thus the drift compares the available exposure slack with estimation noise. At a root- T distance, sample fluctuations can still determine whether the constraint binds, even though the population target is feasible.

Define the exposure derivatives by

$$D_\eta \mathfrak{c}_{\alpha,T}(0_d, Q_{Z,T})[v] = \gamma'_{\eta,T} v, \quad D_Q \mathfrak{c}_{\alpha,T}(0_d, Q_{Z,T})[\Delta Q] = \gamma'_{Q,T} \text{vec}(\Delta Q). \quad (7.2)$$

Here $v \in \mathbb{R}^d$ is a candidate coordinate perturbation and $\Delta Q \in \mathbb{S}^L$ is a symmetric metric perturbation. The vector $\gamma_{\eta,T} \in \mathbb{R}^d$ is the boundary normal in local coordinates, and $\gamma_{Q,T} \in \mathbb{R}^{L^2}$ measures its metric sensitivity. The chain rule starts from $Db_\alpha[\dot{\Gamma}_\alpha, \Delta Q] = 2\dot{\Gamma}'_\alpha Q \dot{\Gamma}_\alpha + \dot{\Gamma}'_\alpha \Delta Q \dot{\Gamma}_\alpha$. The second derivative in (7.2) therefore includes the response of the metric-selected representative as well as the direct quadratic-form term.

Set $v = \sqrt{T} \eta$, $\mathcal{Z}_T = T^{-1/2} \sum_t \xi_{t,P,T}$, and $\mathcal{Z}_{Q,T} = \sqrt{T} \text{vec}(\widehat{Q}_Z - Q_{Z,T})$. Here $\xi_{t,P,T}$ is the complete score for that law and its unrestricted target. Let

$$G_T(v) = T\{\widetilde{Q}_T(v/\sqrt{T}; \widehat{Q}_Z) - \widetilde{Q}_T(0_d; \widehat{Q}_Z)\}.$$

Probability orders below are uniform over $\mathfrak{P}_T$: $R_T = o_p(1)$ means $\sup_{P \in \mathfrak{P}_T} P(\|R_T\| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$; $O_p(1)$ means uniform tightness.

Assumption 8 (Uniform boundary experiment). *There are fixed $H \succ 0$, $\gamma_{\eta,0} \neq 0_d$, $\gamma_{Q,0}$, and*

$\mathcal{J}_{\xi Q} \succeq 0$ such that:

(i) For every finite M ,

$$\sup_{\|v\| \leq M} |G_T(v) - v' \mathcal{Z}_T - \frac{1}{2} v' H v| = o_p(1), \quad (7.3)$$

$$\sup_{\|v\| \leq M} |\sqrt{T} \mathfrak{c}_{\alpha, T}(v/\sqrt{T}, \widehat{Q}_Z) - \kappa_T - \gamma'_{\eta, 0} v - \gamma'_{Q, 0} \mathcal{Z}_{Q, T}| = o_p(1). \quad (7.4)$$

- (ii) The joint vector $(\mathcal{Z}'_T, \mathcal{Z}'_{Q, T})'$ converges uniformly in bounded-Lipschitz distance to $(\mathcal{Z}', \mathcal{Z}'_Q)' \sim \mathcal{N}(0, \mathcal{J}_{\xi Q})$. The test functions in this distance are bounded by one and have Lipschitz constant at most one.
- (iii) The selected estimate $\widehat{v}_T = \sqrt{T} \widehat{\eta}_T$ is $O_p(1)$, is feasible with probability tending to one uniformly, and achieves an $o_p(1)$ gap in G_T relative to the infimum over the sample-feasible chart domain.
- (iv) For the scalar report ϕ_T , its derivatives at $(0_d, Q_{Z, T})$ converge uniformly to $\phi_{\eta, 0}$ and $\phi_{Q, 0}$. On common local neighborhoods the derivatives are bounded and have a common modulus of continuity tending to zero with the argument distance.

The fixed radius M restricts the expansions to bounded root- T movements, not the entire parameter space. Tightness in part (iii) permits this localization. Feasibility and branch selection are requirements on the chosen estimator, not consequences of a score expansion. Uniformity makes these approximations valid for nearby laws whose exposure slack is not known to the researcher. A pointwise approximation at a single fixed slack would not provide that transition guarantee.

The expansions can be checked using the preceding profile derivatives. Specifically, suppose the uniform derivative laws, score tightness, and smooth chart/profile conditions hold with common constants over the family; the finite- T Hessians and exposure derivatives converge uniformly to the displayed limits; and the exposure has uniformly bounded second derivatives. The integral Taylor formula gives (7.3). For the constraint its scaled remainder on $\|v\| \leq M$ is bounded by

$$O_p\{T^{-1/2}(M + \|\mathcal{Z}_{Q, T}\|)^2\} = o_p(1).$$

Replacing the derivatives by their limits adds at most

$$M\|\gamma_{\eta, T} - \gamma_{\eta, 0}\| + \|\gamma_{Q, T} - \gamma_{Q, 0}\| \|\mathcal{Z}_{Q, T}\| = o_p(1).$$

This verifies part (i). It does not derive the uniform CLT or the estimator localization from smoothness alone. The boundary experiment replaces, rather than imposes, the interior strict-slackness condition.

7.2 Projection and the scalar report

Along any selection with $\kappa_T \rightarrow \kappa \in \bar{\mathbb{K}} := [-\infty, 0]$, define for finite κ the random half-space

$$\mathcal{T}_\kappa(\mathcal{Z}_Q) = \{\nu : \gamma'_{\eta,0}\nu + \gamma'_{Q,0}\mathcal{Z}_Q + \kappa \leq 0\}. \quad (7.5)$$

Its defining expression is the first-order constraint excess: the candidate loading movement, the learned-metric shift, and population local slack. Set $\mathcal{T}_{-\infty} = \mathbb{R}^d$.

Theorem 3 (Learned-boundary projection and report limit). *Under Assumption 8, along the stated selection, $\widehat{\nu}_T \Rightarrow \nu_\kappa$. For finite κ ,*

$$\begin{aligned} \nu^\mu &= -H^{-1}\mathcal{Z}, & d_0 &= \gamma'_{\eta,0}H^{-1}\gamma_{\eta,0} > 0, \\ \nu_\kappa &= \nu^\mu - H^{-1}\gamma_{\eta,0} \frac{[\gamma'_{\eta,0}\nu^\mu + \gamma'_{Q,0}\mathcal{Z}_Q + \kappa]_+}{d_0}, \end{aligned} \quad (7.6)$$

where $[x]_+ = \max(x, 0)$, and $\nu_{-\infty} = \nu^\mu$. For $\widehat{\phi}_T = \phi_T(\widehat{\eta}_T, \widehat{Q}_Z)$ and $\phi_{0,T} = \phi_T(0_d, Q_{Z,T})$,

$$\sqrt{T}(\widehat{\phi}_T - \phi_{0,T}) \Rightarrow L_\phi(\kappa) := \phi_{\eta,0}\nu_\kappa + \phi_{Q,0}\mathcal{Z}_Q. \quad (7.7)$$

Proof. Fix a convergent drift selection. For finite κ , the constraint expansion gives local convergence of feasible sets to (7.5). In detail, every limit of feasible bounded sequences satisfies the limiting inequality. A strictly feasible limiting point is eventually feasible under uniform approximation. A point on the limiting boundary can be approximated by strictly feasible points $\nu - \varepsilon\gamma_{\eta,0}$, because $\gamma_{\eta,0} \neq 0$. These are the outer and inner approximations needed for constrained minimization. Uniform convergence of the objective on bounded sets, approximate optimality, and tightness then give the unique minimizer of $\nu'\mathcal{Z} + \frac{1}{2}\nu'H\nu$ over this half-space. This is the localized constrained argmin argument (Geyer, 1994; Andrews, 1999; Shapiro, 2000). At $\kappa = -\infty$, tightness of the shocks makes every fixed ball eventually feasible with probability tending to one, giving the unrestricted minimizer instead.

To solve the finite-drift problem, complete the square:

$$\nu'\mathcal{Z} + \frac{1}{2}\nu'H\nu = \frac{1}{2}(\nu - \nu^\mu)'H(\nu - \nu^\mu) - \frac{1}{2}(\nu^\mu)'H\nu^\mu.$$

The last term is constant in ν , so this is an H -metric projection. The first-order condition for its Lagrangian is $\nu = \nu'' - \lambda H^{-1} \gamma_{\eta,0}$ with $\lambda \geq 0$. If the unrestricted movement is feasible, $\lambda = 0$. Otherwise the boundary equality gives $\lambda = (\gamma'_{\eta,0} \nu'' + \gamma'_{Q,0} \mathcal{Z}_Q + \kappa)/d_0$. Combining the cases proves (7.6).

Finally, the integral first-order formula for the report gives

$$\sqrt{T}(\widehat{\phi}_T - \phi_{0,T}) = \phi_{\eta,0} \widehat{\nu}_T + \phi_{Q,0} \mathcal{Z}_{Q,T} + o_p(1). \quad (7.8)$$

To bound its remainder, the joint displacement is $O_p(T^{-1/2})$. Its root- T norm is tight and multiplies a derivative modulus tending uniformly to zero by part (iv). Derivative replacement is also $o_p(1)$ by tightness. The continuous mapping theorem completes the report limit. $\square$

Metric estimation enters ν'' through the complete score and shifts the half-space through $\gamma'_{Q,0} \mathcal{Z}_Q$. The report can have a third, direct metric response. The unrestricted movement is retained when it respects the learned budget; otherwise it is moved back to the boundary in the direction determined by curvature. This explains why an exposure restriction can produce an asymmetric sampling law without changing the Gaussian input assumption. For a report of residual intercept exposure or benchmark alignment, the relevant uncertainty is therefore the projected movement, not simply a larger symmetric standard error.

7.3 Feasible simulation and endpoint envelopes

Estimate the long-run covariance of $\widehat{y}_t = (\widehat{\xi}'_t, \widehat{\varphi}'_{Q,t})'$ using the centered HAC construction in (6.9). Conditional on the data, draw independent $(\mathcal{Z}^{(b)'}_t, \mathcal{Z}^{(b)'}_Q)' \sim \mathcal{N}(0, \widehat{\mathcal{J}}_{\xi Q})$. Apply (7.6) with plug-in derivatives and then (7.7). Use the same draws at every drift.

This simulates the estimated first-order experiment, not a resampled raw panel. A positive-semidefinite covariance suffices: full vectorization of symmetric matrices naturally gives redundant coordinates, and a singular Gaussian can be simulated on its covariance range. Symmetrization and clipping negative eigenvalues at zero preserve consistency. A positive numerical floor must vanish asymptotically if it affects zero population eigenvalues.

Let $q_\tau(\kappa)$ denote the lower τ quantile of $L_\phi(\kappa)$. For a nominal error probability $\alpha \in (0, 1)$, define

$$q_\alpha^L = \inf_{\kappa \in \mathbb{K}} q_{\alpha/2}(\kappa), \quad q_\alpha^U = \sup_{\kappa \in \mathbb{K}} q_{1-\alpha/2}(\kappa). \quad (7.9)$$

These are least-favorable tail envelopes over the unknown local slack, in the sense of uniform

asymptotic inference (Andrews and Guggenberger, 2009; Andrews and Soares, 2010). In this particular scalar, single-constraint experiment, only two drifts are needed.

Proposition 6 (Endpoint sufficiency). *For the law in (7.7),*

$$q_\alpha^L = \min_{\kappa \in \{-\infty, 0\}} q_{\alpha/2}(\kappa), \quad q_\alpha^U = \max_{\kappa \in \{-\infty, 0\}} q_{1-\alpha/2}(\kappa). \quad (7.10)$$

The same identity holds for empirical quantiles computed with common Gaussian draws and fixed plug-in derivatives.

Proof. Write $A = \gamma'_{\eta,0} \nu'' + \gamma'_{Q,0} \mathcal{Z}_Q$, $U = \phi_{\eta,0} \nu'' + \phi_{Q,0} \mathcal{Z}_Q$, and $c = \phi_{\eta,0} H^{-1} \gamma_{\eta,0} / d_0$. Then $L_\phi(\kappa) = U - c[A + \kappa]_+$. Every sample path is nonincreasing in κ if $c \geq 0$ and nondecreasing if $c < 0$. Pathwise ordering implies the same ordering of all quantiles, with endpoints $-\infty$ and zero. The identical argument applies to a finite common-draw sample. $\square$

Thus no growing drift grid is required for the stated theorem. The reduction makes the procedure directly implementable: simulate one joint Gaussian sample, apply the interior and binding maps, and retain the more protective critical value in each tail. Reusing the draws also prevents a comparison across drifts from being driven by independent Monte Carlo noise. The reduction uses a scalar report, one half-space, and drift-independent limit derivatives; it does not extend automatically to multiple inequalities or vector confidence regions.

Assumption 9 (Feasible boundary inference). *Uniformly over $\mathfrak{P}_T$:*

- (i) *The covariance and derivative estimators used in simulation are consistent for $\mathcal{J}_{\xi Q}$ and $(H, \gamma_{\eta,0}, \gamma_{Q,0}, \phi_{\eta,0}, \phi_{Q,0})$. The simulated covariance is positive semidefinite, and any numerical covariance adjustment is $o_p(1)$. With probability tending to one, $\widehat{H} \succ 0$ and $\widehat{\gamma}'_\eta \widehat{H}^{-1} \widehat{\gamma}_\eta$ is bounded away from zero.*
- (ii) *In common neighborhoods of its $\alpha/2$ and $1 - \alpha/2$ quantiles, each $L_\phi(\kappa)$ has a density bounded above and away from zero, uniformly in $\kappa \in \overline{\mathbb{K}}$.*
- (iii) *The number N_T^{sim} of conditional independent Gaussian draws tends to infinity.*

The local density condition concerns the critical quantiles, not the entire distribution. A projection can create atoms; a critical quantile lying at an atom is outside this maintained condition. In particular, at exact binding the exposure report itself has limit $\min(\gamma'_{\eta,0} \nu'' + \gamma'_{Q,0} \mathcal{Z}_Q, 0)$. Its upper critical quantile can be an atom at zero; the interval theorem below

then does not apply to that report. HAC conditions such as Assumption 7, imposed uniformly for the stacked influences, supply the covariance requirement.

Let $\widehat{q}_\alpha^L$ and $\widehat{q}_\alpha^U$ be the two-endpoint empirical versions of (7.10). The drift-robust interval is

$$\text{CI}_{1-\alpha}^{\text{DR}} = \left[\widehat{\phi}_T - \frac{\widehat{q}_\alpha^U}{\sqrt{T}}, \widehat{\phi}_T - \frac{\widehat{q}_\alpha^L}{\sqrt{T}} \right]. \quad (7.11)$$

Theorem 4 (Uniform drift-robust coverage). *Under Assumptions 8 and 9, the estimated envelope endpoints converge uniformly in probability to their population values, and*

$$\liminf_{T \rightarrow \infty} \inf_{P \in \mathfrak{P}_T} P\{\phi_{0,P,T} \in \text{CI}_{1-\alpha}^{\text{DR}}\} \geq 1 - \alpha. \quad (7.12)$$

Proof. Use a common standard Gaussian vector to represent each plug-in draw through the symmetric square root of its covariance. The square-root map is continuous on the positive-semidefinite cone, including singular matrices. The projection and report maps are continuous when the curvature and normal denominator remain regular. Uniform consistency of their inputs therefore gives convergence of the two endpoint laws. The local density bound gives convergence of their critical quantiles. Conditional empirical distribution bounds for independent draws make the Monte Carlo errors negligible as the number of draws diverges. Taking a minimum or maximum of two consistent quantiles proves envelope consistency.

For coverage, each drift satisfies $q_\alpha^L \leq q_{\alpha/2}(\kappa)$ and $q_\alpha^U \geq q_{1-\alpha/2}(\kappa)$. Suppose uniform coverage failed. There would be an $\varepsilon > 0$, a subsequence T_j , and laws $P_j \in \mathfrak{P}_{T_j}$ with coverage less than $1 - \alpha - \varepsilon$. The set $[-\infty, 0]$ is compact under $d(\kappa_1, \kappa_2) = |e^{\kappa_1} - e^{\kappa_2}|$, $e^{-\infty} = 0$. Pass to a further subsequence with $\kappa_{T_j}(P_j) \rightarrow \kappa^*$. The uniform experiment and report expansion yield $\sqrt{T_j}(\widehat{\phi}_{T_j} - \phi_{0,P_j,T_j}) \Rightarrow L_\phi(\kappa^*)$ on this selection.

For any sufficiently small $h > 0$, the interval $(q_{\alpha/2}(\kappa^*) + h, q_{1-\alpha/2}(\kappa^*) - h)$ is contained in the estimated envelope interval with probability tending to one. The open-set convergence bound gives limiting lower coverage at least the probability of this inner interval under $L_\phi(\kappa^*)$. Letting $h \downarrow 0$ gives $1 - \alpha$ by continuity at the two pointwise quantiles, contradicting the selected undercoverage. This argument does not require continuity at every envelope endpoint. $\square$

The interval can be conservative because different endpoints can determine its two tails. If exact binding is known from the model, the pointwise zero-drift interval is weakly shorter. Without that information, substituting zero for unknown local slack does not give the uniform guarantee above. The theorem supplies a coverage guarantee when the analyst cannot tell

reliably whether an exposure budget is effectively slack. Its practical cost is a potentially wider interval, not a change in the estimated loading class. Table III and Figure 3 quantify that coverage–length tradeoff in the local experiment.

7.4 Scope

A zero budget with $Q \succ 0$ imposes $\Gamma_\alpha = 0$ exactly. Its quadratic constraint has zero normal at zero, so it is not a smooth half-space problem. If instead $\Gamma_{\alpha,0} = 0$ with a positive slack budget, the loading estimator may remain regular while exposure is second order: if $\sqrt{T} \hat{\Gamma}_\alpha \Rightarrow \mathcal{Z}_\alpha$ and $\hat{Q}_Z \rightarrow_p Q_Z$, then

$$T \hat{\Gamma}_\alpha' \hat{Q}_Z \hat{\Gamma}_\alpha \Rightarrow \mathcal{Z}_\alpha' Q_Z \mathcal{Z}_\alpha. \quad (7.13)$$

Indeed the metric-error term is bounded by $\|\sqrt{T} \hat{\Gamma}_\alpha\|^2 \|\hat{Q}_Z - Q_Z\| = o_p(1)$.

Nonsmooth spectral reports require their own regularity analysis; a directional delta method applies only after directional differentiability has been established (Fang and Santos, 2019). Weak curvature or loss of profile rank likewise falls outside the maintained local separation conditions (Stock and Wright, 2000; Andrews and Cheng, 2012). The present result concerns a single smooth learned boundary, not arbitrary multiple constraints or the validity of an unverified alternating algorithm.

8 Calibration and Simulation Evidence

The numerical analysis asks whether the distinctions in the theory matter for measured exposures and their uncertainty. The empirical calibration first shows why the geometry of recorded characteristics is consequential. The factor-panel experiment then compares inference methods at one point estimate. Finally, a local experiment isolates the additional effect of learning an exposure boundary. Keeping these comparisons separate lets each exhibit identify the mechanism it is designed to assess.

8.1 Characteristic geometry

The calibration uses an assembled U.S. stock panel with 153 monthly characteristics, January 1962–December 2025, with definitions following Jensen et al. (2023). Within each month, percentile ranks are rescaled to $[-1/2, 1/2]$. Missing observations receive zero; a

column with fewer than ten valid observations is set to zero for that month. The aggregate Gram matrix averages monthly cross-sectional second moments. Trace normalization,

$$Q^{\text{cal}} = L\widehat{Q}_Z/\text{tr}(\widehat{Q}_Z), \quad (8.1)$$

sets the average eigenvalue to one without changing spectral shares, orthogonality, or the condition number. Thus its comparison with I_{153} holds the overall scale fixed while changing relative lengths and overlap.

Table I: Characteristic-space geometry

Diagnostic	Q^{cal}	I_{153}
Leading spectral share	0.132	0.007
Share of first five directions	0.379	0.033
Directions for 90% of mass	68	138
Mean absolute off-diagonal entry	0.114	0.000
Maximum/minimum eigenvalue	2,339	1

Notes: The retained aggregate Gram matrix summarizes 153 monthly ranked characteristics. Trace normalization sets the average eigenvalue to one. Spectral shares divide eigenvalues by their sum; the identity assigns equal mass to all coordinate directions. The input spans January 1962–December 2025. The replication package records the matrix and preprocessing.

Table I summarizes that difference. The leading direction carries 13.2 percent of spectral mass, the first five carry 37.9 percent, and 68 directions account for 90 percent. The identity metric instead requires 138. Figure 1 shows both the cumulative spectrum and the overlap between individual characteristic directions. For two coordinate directions $j \neq k$, the latter is measured by $|Q_{jk}^{\text{cal}}|/(Q_{jj}^{\text{cal}}Q_{kk}^{\text{cal}})^{1/2}$. It is an absolute cosine under the Gram inner product, not necessarily a centered correlation.

The median absolute overlap is 0.060, and its 95th percentile is 0.380. The implication is not that every signal is nearly redundant. Rather, individual scaling leaves a nontrivial set of overlapping exposures and a concentrated aggregate spectrum. For an investor comparing

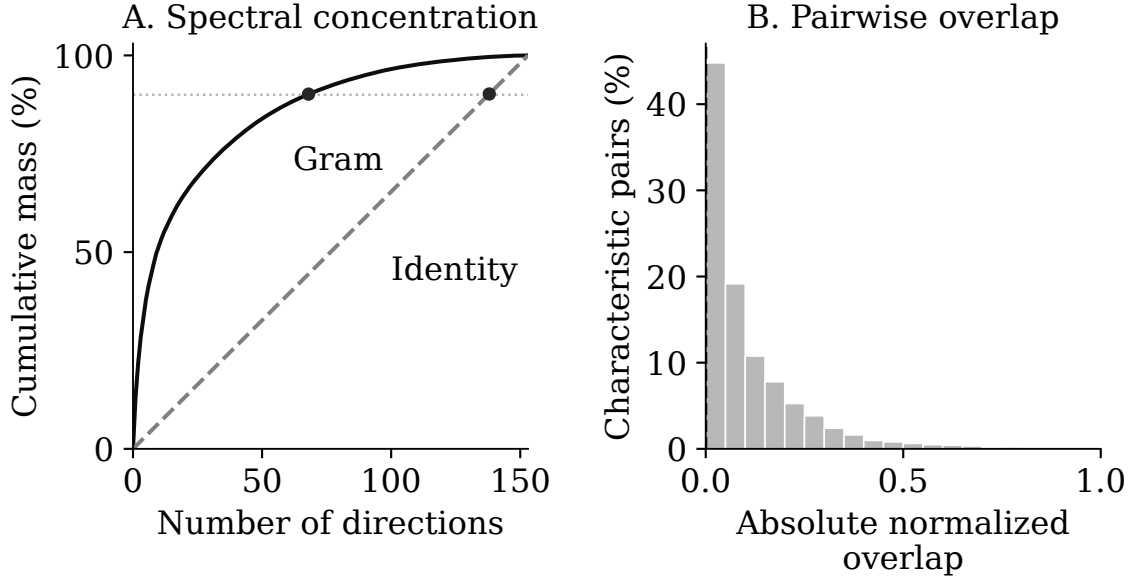


Figure 1: Concentration and overlap in characteristic space. Panel A cumulates descending eigenvalues divided by their sum; the horizontal line marks 90 percent. Panel B shows the share of all 11,628 distinct characteristic pairs in each absolute-overlap bin. The dashed line at zero is the identity-metric benchmark. Both panels use the same 153-characteristic aggregate Gram matrix; neither uses return outcomes.

value, profitability, or related signals, different coefficient directions can still place exposure on similar stocks. The Gram metric records this overlap when loading blocks are separated. This is the empirical setting for Proposition 1: the allocation can depend on the ruler even when a matched transformation preserves fitted returns. The spectrum measures characteristic exposure, not the number of priced risks.

8.2 The profiled-estimator experiment

We next ask how much uncertainty is missed when an estimated metric is treated as known. The baseline has $L = 12$, $K = 2$, $K_\phi = 1$, $N = 100$, and equal date weights. Characteristics are Gaussian with a trace-normalized population covariance of condition number eight. Latent factors are stationary AR(1) processes with coefficient 0.6, the observed factor has correlation 0.6 with the first latent factor, and unit-level disturbances are conditionally spherical with standard deviation 0.6. These controlled choices introduce

anisotropy and factor persistence while keeping the population metric well conditioned. They are not estimates of a unique empirical data-generating process.

Each sample estimates its metric. Four alternating updates initialize zero-ridge quotient polishing to a root-mean-square score tolerance of 10^{-8} . The exposure budget is five; all 4,000 recorded estimates pass the numerical gates and remain strictly below the budget. The reports are $a'P_Q(B)a$ for fixed unit vectors a in whitened coordinates. A separate calibration selected moderate- and high-sensitivity directions from sixteen candidates, both with population value 0.5. This calibration is separate from the coverage draws. At each $T \in \{240, 960\}$, 2,000 replications use the same point estimate for every inference method. The Online Supplement gives the full design, frozen selection rule, and numerical tolerances.

Table II separates point-estimator dispersion from standard-error estimation. The Monte Carlo scale uses the empirical standard deviation across replications; oracle-influence HAC uses structural loadings and derivatives frozen from the independent long calibration. The three feasible rows respectively omit metric uncertainty, include its marginal variance without the cross term, and retain the complete influence. The ratio SE/SD should be near one, and nominal 95 percent intervals should cover in approximately 95 percent of draws.

At $T = 960$, fixed-metric standard errors are only 83.8 and 89.1 percent of the Monte Carlo standard deviations for the two reports. Complete-score HAC raises these ratios to 99.6 and 101.6 percent. Coverage rises from 89.05 to 94.25 percent and from 91.75 to 94.85 percent. The paired gains are 5.20 and 3.10 percentage points, with Monte Carlo standard errors of 0.50 and 0.39 points. These paired comparisons use the same simulated panel for both methods; the difference is uncertainty measurement, not a different estimate of the loading class.

Figure 2 explains the coverage difference. When the metric is treated as fixed, both tails lie farther from zero than the normal benchmark: an ordinary estimation error appears too large relative to its reported standard error. The complete-score curves are closer to the diagonal. For a researcher reporting alignment with a benchmark exposure, the correction changes the precision of that statement without changing the estimated alignment itself.

The no-cross and complete rows are close in this design. The supplementary variance decomposition attributes about 29 and 23 percent of complete variance to the metric component at $T = 960$, while the signed cross terms are close to zero. Thus these simulations primarily demonstrate omitted metric variance, not a large cross-covariance effect. At $T = 240$, feasible and oracle-influence HAC both under-cover modestly; correcting the influence does not eliminate finite-sample covariance-estimation error.

Table II: Standard-error calibration and 95 percent coverage

Method	$T = 240$		$T = 960$	
	SE/SD	Coverage	SE/SD	Coverage
Panel A: Moderate-sensitivity report				
Monte Carlo scale	1.000	94.90	1.000	94.75
Oracle-influence HAC	0.985	94.15	1.004	94.90
Metric treated as known	0.820	88.20	0.838	89.05
Metric, no cross term	0.978	93.95	0.997	94.65
Complete-score HAC	0.976	93.75	0.996	94.25
Panel B: High-sensitivity report				
Monte Carlo scale	1.000	95.25	1.000	95.10
Oracle-influence HAC	0.957	93.60	1.037	95.20
Metric treated as known	0.829	88.15	0.891	91.75
Metric, no cross term	0.948	93.40	1.017	95.05
Complete-score HAC	0.942	93.20	1.016	94.85

Notes: Each horizon uses 2,000 panels with $L = 12$, $K = 2$, $K_o = 1$, $N = 100$, and one polished point estimate across methods. SE/SD divides mean estimated standard error by the Monte Carlo standard deviation; its benchmark is one. Coverage is in percent, with a 95 percent benchmark. Both reports have population value 0.5. Oracle HAC uses structural loadings and reference derivatives frozen from a separate long calibration. The fixed-metric calculation omits metric uncertainty; the no-cross row adds metric variance but omits the signed cross-covariance. Coverage Monte Carlo standard error near 95 percent is 0.5 percentage points.

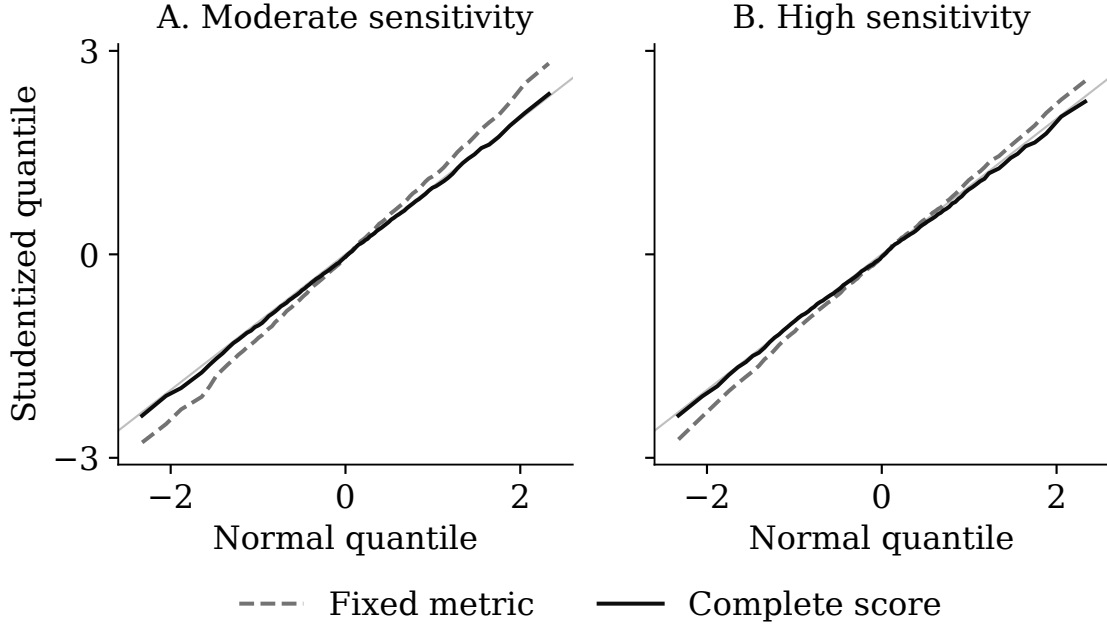


Figure 2: Distributional calibration of the two projector reports. At $T = 960$, each curve plots the empirical quantiles of $(\hat{\phi} - 0.5)/\hat{se}$ against standard-normal quantiles, using 2,000 replications and probabilities from 0.01 to 0.99. The light diagonal is the normal benchmark. Point estimates are identical across the two standard-error methods.

The experiment evaluates the estimator under a Gaussian fixed- N design. Its datewise full profile rank does not imply the stronger all-date lower bound in (4.13). Conditional sphericity centers the structural score, and the numerical gates check the realized solutions; an inverse-moment argument would be needed to place this DGP under a weaker primitive theorem. The evidence therefore concerns the finite-sample uncertainty mechanism, with that distinction explicit.

8.3 The learned-boundary experiment

The final comparison asks what changes when the same estimated metric also moves an exposure restriction. It implements the local experiment of Theorem 3. A stationary correlated process supplies a two-dimensional complete score and a scalar metric influence. Curvature is positive, and both the boundary and report respond to the metric. The evaluation drifts are $\kappa \in \{-4, -2, -1, -0.5, 0\}$: zero is binding, and more negative values allow greater

population slack relative to root- T sampling error. Every comparison holds the point estimate fixed.

The five interval rules isolate different information requirements. No projection uses the interior Gaussian law. Fixed boundary knows the drift but suppresses the metric-induced boundary shift. Binding only always calibrates at zero. Known drift retains the learned boundary and the true drift, providing an infeasible reference. Drift robust instead protects over the unknown drift using the proposed envelope.

Table III: Learned-boundary coverage at $T = 960$

κ	No projection	Fixed boundary	Binding only	Known drift	Drift robust
Nominal 95% intervals					
-4	93.20	93.20	83.80	92.90	96.30
-2	94.95	93.85	87.35	93.30	98.20
-1	94.90	92.95	92.20	93.40	98.75
-0.5	93.15	91.80	94.30	94.35	98.45
0	90.15	91.20	94.70	94.70	97.10
Length at $\kappa = 0$	2.775	2.261	2.584	2.584	3.241

Notes: Coverage is in percent, from 2,000 replications and 1,499 Gaussian draws. Length is mean interval length multiplied by $\sqrt{T}$, evaluated only at $\kappa = 0$. No projection uses a Gaussian interval; Fixed boundary knows the drift but omits the metric boundary shift; Binding only calibrates at zero; Known drift retains the learned boundary and true drift; Drift robust protects over unknown drift. All methods use one point estimator. Coverage Monte Carlo standard error near 95 percent is 0.5 percentage points.

At $T = 960$ and exact binding, Table III shows coverage of 90.15 percent without projection and 91.20 percent when the boundary shift is omitted. Binding-only calibration covers 94.70 percent at binding but only 83.80 percent at $\kappa = -4$. An estimated binding constraint therefore cannot simply be used to justify zero-drift critical values for all nearby population targets. Drift-robust coverage ranges from 96.30 to 98.75 percent across the five

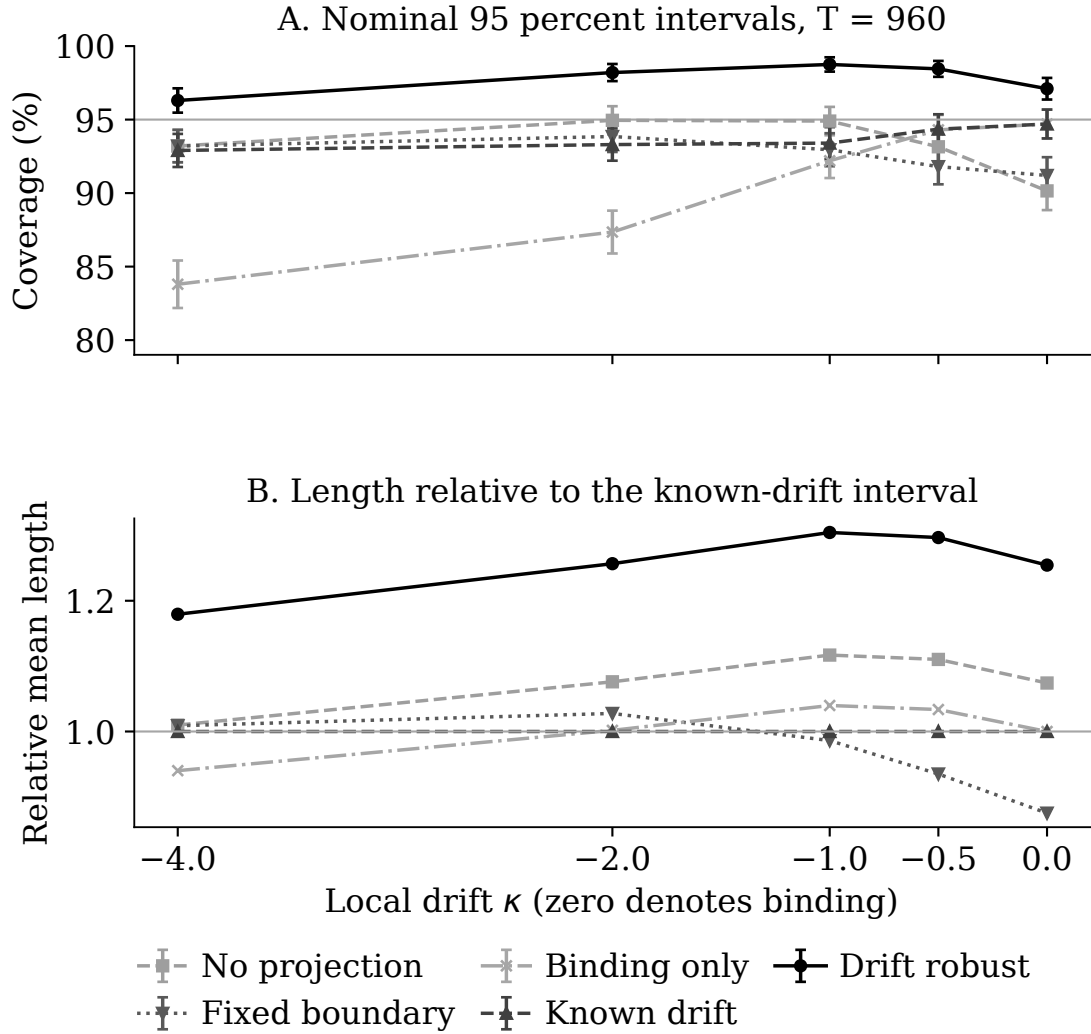


Figure 3: Coverage and the cost of protecting against unknown slack. Panel A reports 95 percent coverage at $T = 960$; vertical bars are pointwise Monte Carlo intervals, $\hat{p} \pm 1.96\sqrt{\hat{p}(1 - \hat{p})/2000}$. Panel B divides each method's mean interval length by the known-drift mean length at the same drift. All methods use the same local estimator, 2,000 replications, and 1,499 Gaussian draws. Lines join the five evaluated drifts; the intervals are not simultaneous bands.

drifts.

Figure 3 makes the tradeoff visible. Protecting against unknown slack raises coverage and requires a longer interval than knowing the drift. At binding, the mean root- T lengths are 3.241 and 2.584, respectively, a difference of about 25 percent. For an exposure-budget

application, this is the precision cost of reporting an interval without first classifying the population target as slack or binding. It is not a gain in precision or a change in the fitted model.

The archived implementation envelopes a grid and the interior endpoint. Proposition 6 proves that common draws and fixed derivatives reduce this calculation to two endpoints; the replication package checks the equality numerically. Additional results at $T = 240$ show the same comparison with a shorter sample. This experiment isolates the joint score–boundary–report law rather than re-estimating a raw factor panel under a binding budget. Together with the interior experiment, it locates two distinct sources of inferential error: omitting how the ruler is learned, and omitting how it moves the feasible set.

9 Conclusion

Characteristic-based factor decompositions depend on the geometry used to normalize and separate loading blocks. Making that geometry explicit distinguishes invariant fitted content from metric-dependent attribution. It also identifies which empirical comparisons require uncertainty from learning the metric.

For an exactly profiled estimator, the complete score combines return and metric variation, and smooth reports translate its quotient law into chart-invariant inference. Covariance alignment supplies a sufficient bridge from the criterion target to structural loadings without requiring the cross section to grow. Near a smooth exposure boundary, the same learned metric moves the objective and feasible set. Their joint projection law supports drift-robust scalar intervals; for one constraint their tail envelopes reduce to two endpoint calculations.

The analysis is local, finite dimensional, and strongly identified. Inverse-moment profile conditions, increasing characteristic dimension, multiple constraints, and weak-curvature procedures are substantive extensions. The present framework supplies a common target, derivative, and covariance language in which those extensions can be developed and empirical attribution can be reported reproducibly.

Data and Code Availability

The replication supplement supplies simulation programs, frozen designs, archived records, and exhibit-rebuilding code. It distinguishes archived reconstruction from fresh estimation. The empirical Gram matrix can be rebuilt from the recorded annual stock-

characteristic files; this requires the same input vintage, and stock-level data are not bundled. The availability statement documents data access and code redistribution requirements. This working-paper package has no public archive identifier.

References

- Pierre-Antoine Absil, Robert Mahony, and Rodolphe Sepulchre. *Optimization Algorithms on Matrix Manifolds*. Princeton University Press, 2008.
- Donald W. K. Andrews. Heteroskedasticity and autocorrelation consistent covariance matrix estimation. *Econometrica*, 59(3):817–858, 1991.
- Donald W. K. Andrews. Estimation when a parameter is on a boundary. *Econometrica*, 67(6):1341–1383, 1999.
- Donald W. K. Andrews and Xu Cheng. Estimation and inference with weak, semi-strong, and strong identification. *Econometrica*, 80(5):2153–2211, 2012.
- Donald W. K. Andrews and Patrik Guggenberger. Validity of subsampling and “plug-in asymptotic” inference for parameters defined by moment inequalities. *Econometric Theory*, 25(3):669–709, 2009. doi: 10.1017/S0266466608090257.
- Donald W. K. Andrews and Gustavo Soares. Inference for parameters defined by moment inequalities using generalized moment selection. *Econometrica*, 78(1):119–157, 2010. doi: 10.3982/ECTA7502.
- Jushan Bai. Inferential theory for factor models of large dimensions. *Econometrica*, 71(1): 135–171, 2003.
- Jushan Bai and Serena Ng. Principal components estimation and identification of static factors. *Journal of Econometrics*, 176(1):18–29, 2013.
- Alan Edelman, Tomás A. Arias, and Steven T. Smith. The geometry of algorithms with orthogonality constraints. *SIAM Journal on Matrix Analysis and Applications*, 20(2): 303–353, 1998.
- Zheng Fang and Andres Santos. Inference on directionally differentiable functions. *The Review of Economic Studies*, 86(1):377–412, 2019.
- Charles J. Geyer. On the asymptotics of constrained M-estimation. *The Annals of Statistics*, 22(4):1993–2010, 1994.
- Theis Ingerslev Jensen, Bryan T. Kelly, and Lasse Heje Pedersen. Is there a replication crisis in finance? *Journal of Finance*, 78(5):2465–2518, 2023.

- Bryan T. Kelly, Seth Pruitt, and Yinan Su. Characteristics are covariances: A unified model of risk and return. *Journal of Financial Economics*, 134(3):501–524, 2019.
- Bryan T. Kelly, Sofonias A. Korsaye, Seth Pruitt, and Yinan Su. Instrumented principal component analysis. Technical report, SSRN Working Paper 2983919, 2026. Working paper dated April 21, 2026.
- Jessie Li. The proximal bootstrap for constrained estimators. *Journal of Statistical Planning and Inference*, 236:106245, 2025. doi: 10.1016/j.jspi.2024.106245.
- Whitney K. Newey. The asymptotic variance of semiparametric estimators. *Econometrica*, 62(6):1349–1382, 1994.
- Whitney K. Newey and Daniel McFadden. Large sample estimation and hypothesis testing. In Robert F. Engle and Daniel L. McFadden, editors, *Handbook of Econometrics*, volume 4, pages 2111–2245. Elsevier, Amsterdam, 1994.
- Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.
- Alexander Shapiro. On the asymptotics of constrained local M-estimators. *The Annals of Statistics*, 28(3):948–960, 2000.
- James H. Stock and Jonathan H. Wright. GMM with weak identification. *Econometrica*, 68(5):1055–1096, 2000.
- Paolo Zaffaroni. Factor models for conditional asset pricing. *Journal of Political Economy*, 133(8):2615–2642, 2025.